AKSHIT JAIN

Machine Learning Engineer | Full-Stack Developer | AI Systems Builder

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SKILLS

Languages:	Python, C, C++, Java, JavaScript
Machine Learning:	Natural Language Processing (NLP), OCR-based Systems, Healthcare AI, Classification & Regression Models, Clustering (K-Modes), Model Training Pipelines, Data Processing
Frameworks & Libraries:	Django, Flask, React, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib
Tools/Platforms:	Git, GitHub, Streamlit, Vercel, TensorFlow, Jupyter Notebook, VS Code, MySQL, SQLite, Tesseract OCR, Gemini API, Gmail SMTP, Chart.js, pyttsx3, googletrans, tkinter
Web Development:	HTML, CSS, Bootstrap, Tailwind CSS, JavaScript, Responsive UI Design
Core Concepts:	Data Structures & Algorithms, Problem Solving, Optimization Techniques, System Design Basics
Soft Skills:	Teamwork, Leadership, Adaptability, Critical Thinking, Communication

PROJECTS

Explainable Clinical Entity Recognition (EHR NLP System)| *Python, HuggingFace, NLP, Streamlit*| GitHub

- Developed an end-to-end transformer-based NLP pipeline to extract clinical entities from electronic health records (EHR) while ensuring patient data privacy.
- Implemented de-identification techniques with more than 10 PHI masking patterns to anonymize sensitive healthcare information.
- Integrated explainable AI components to provide interpretable model outputs, improving trust and transparency in predictions.
- Achieved approximately 72% model accuracy on biomedical named entity recognition tasks using domain-specific datasets.
- Designed a structured workflow for preprocessing, tokenization, model training, and inference visualization.

Rheumatoid Arthritis Patient Stratification (K-Modes Clustering) | *Python, Pandas, K-Modes, Streamlit, Matplotlib* | GitHub | Live

- Built a machine learning system for clustering rheumatoid arthritis patients using categorical clinical and socioeconomic data.
- Applied K-Modes clustering algorithm to segment patient groups based on comorbidities, lifestyle factors, and demographics.
- Developed clinically interpretable subgroup analysis to assist in personalized healthcare decision-making.
- Integrated visualization tools to present cluster distributions and patterns effectively.
- Contributed to a patent-related methodology focused on structured patient stratification techniques.

Handwritten Assignment Grader | *Python, Streamlit, Tesseract OCR, Gemini API* | GitHub | Live

- Engineered an AI-powered evaluation system to automate grading of handwritten assignments using OCR and large language models.
- Designed a complete pipeline including text extraction, grammar correction, content classification, and rubric-based scoring.
- Integrated Gemini API to generate meaningful feedback and improve evaluation consistency.
- Enabled automated analysis of diverse handwriting styles with improved accuracy.
- Reduced manual grading effort significantly while maintaining structured and unbiased assessment.

Employee Management System| *Django, Python, SQLite, Bootstrap, Gmail SMTP, Chart.js* | GitHub

- Developed a multi-role enterprise-level web application supporting Admin, HR, and Employee dashboards.
- Implemented attendance tracking system with overlap detection and validation mechanisms.
- Built task management module with file attachments and real-time updates.
- Designed leave request and approval workflow with role-based access control.
- Integrated OTP-based authentication and account security features to enhance system reliability

Exam Platform (Django LMS) | *Django, Python, MySQL, Bootstrap, Gmail SMTP*| GitHub

- Designed a secure online examination platform supporting MCQ, subjective, and hybrid assessments.
- Implemented email OTP verification for secure authentication and user validation.
- Developed automated evaluation system for objective questions and structured submission handling.
- Built admin dashboard for exam management, user control, and result tracking.
- Integrated password reset and email notification workflows.

CampusMart (Campus E-commerce Platform) | *Django, Python, MySQL, CSS*| GitHub

- Built a full-stack e-commerce platform tailored for campus users with end-to-end purchase functionality.
- Implemented user authentication, product listing, and dynamic browsing features.
- Designed shopping cart system with order placement and tracking capabilities.
- Developed admin panel for product and order management.
- Optimized database interactions for efficient query handling.

Home Loan Advisor | *Python, Streamlit, Gemini API* | GitHub | Live

- Developed an AI-powered financial advisory application for evaluating home loan eligibility.
- Implemented EMI calculation and loan simulation features for different financial scenarios.
- Integrated conversational AI to provide personalized recommendations based on user inputs.
- Built dynamic interface with real-time updates for improved decision-making.
- Enabled contextual responses for enhanced user interaction experience.

Pathfinding Visualizer | *JavaScript, HTML, Tailwind CSS, Data Structure and Algorithm* | GitHub | Live

- Built an interactive visualization tool to demonstrate pathfinding algorithms including A*, Dijkstra, BFS, and DFS.
- Implemented real-time animation of node traversal and shortest path discovery.
- Added maze generation and grid manipulation features for experimentation.
- Developed algorithm comparison mode to analyze performance differences.
- Enhanced learning experience through step-by-step visual execution.

ByteBazzar (E-commerce Website) | *HTML, CSS, JavaScript*| GitHub | Live

- Designed and developed a responsive e-commerce website with modern UI components.
- Implemented product catalog browsing and structured navigation system.
- Focused on user experience through animations, transitions, and optimized layout design.
- Ensured cross-device compatibility using responsive design techniques.

Eco Tech (E-waste Awareness Platform) | *HTML, CSS, JavaScript, Bootstrap* | GitHub | Live

- Created a responsive web platform to promote awareness about electronic waste and sustainable practices.
- Developed interactive sections with visually engaging content and animations.\
- Implemented Bootstrap-based components for responsive and consistent design.

- Structured content to improve user understanding of environmental impact.

AUTO-NO (Hackathon Project) | *HTML, CSS, JavaScript* | GitHub

- Developed an interactive quiz-based web application during a hackathon environment.
- Implemented dynamic UI components and user interaction logic under time constraints.
- Focused on rapid prototyping and functional delivery within limited time.

Your Buddy Speaker (Desktop Application) | *Python, tkinter, pyttsx3, googletrans* | GitHub

- Built a desktop application for multilingual translation and text-to-speech conversion.
- Implemented real-time translation using googletrans API.
- Integrated speech synthesis functionality using pyttsx3 for audio output.
- Designed user-friendly GUI using tkinter for seamless interaction.
- Enabled voice customization and language selection features.

TRAINING

Hitbullseye | Link June' 25 – Jul' 25

DSA Summer Training (C++)

- Completed structured training covering core Data Structures and Algorithms including arrays, strings, linked lists, stacks, queues, trees, graphs, and dynamic programming.
- Strengthened problem-solving skills by solving curated questions focused on time and space complexity optimization.
- Practiced advanced techniques such as recursion, greedy algorithms, sliding window, and STL-based implementations.
- Improved coding efficiency and logical thinking through consistent problem-solving and revision cycles.

CERTIFICATIONS

- Computational Theory: Language Principle & Finite Automata | **Infosys** | Aug 2025 | [Link]
- Master Generative AI | **Udemy** | Aug 2025 | [Link]
- ChatGPT-4 Prompt Engineering: Generative AI & LLM | **Infosys** | Aug 2025 | [Link]
- Privacy and Security in Online Social Media | **NPTEL** | Oct 2025 | [Link]
- Chat GPT Made Easy | **Udemy** | Aug 2025 | [Link]
- DSA Summer Training | **Hitbullseye** | Jul 2025 | [Link]
- Java Programming | **Lovely Professional University** | May 2025 | [Link]
- Data Structures and Algorithm | **Lovely Professional University** | Dec 2024 | [Link]
- Object Oriented Programming | **Lovely Professional University** | Dec 2024 | [Link]
- Packet Switching Networks and Algorithms | **University of Colorado Boulder** | Nov 2024 | [Link]
- The Bits and Bytes of Computer Networking | **Google** | Sep 2024 | [Link]
- Introduction to Hardware and Operating Systems | **IBM** | Sep 2024 | [Link]
- Mastering Full Stack Development | **Udemy** | Feb 2024 | [Link]
- Digital Systems: From Logic Gates to Processors | **Universitat Autònoma de Barcelona** | Sep 2024 | [Link]
- Computer Programming | **Lovely Professional University** | May 2024 | [Link]
- Hackathon Participation IOT | **IOT Hackathon** | Feb 2024 | [Link]
- Build Generative AI App & Solution (No Code) | **Google Cloud** | Mar 2024 | [Link]
- Pitching Yourself for Opportunities | **LinkedIn** | 2024 | [Link]
- Management Tips | **LinkedIn** | 2024 | [Link]
- Fundamentals of Network | **University of Colorado Boulder** | 2024 | [Link]
- Responsive Web Design | **freeCodeCamp** | Nov 2023 | [Link]

ACHIEVEMENTS

- Contributed to a patent-related methodology in Rheumatoid Arthritis Patient Stratification using K-Modes clustering for categorical medical data.
- Built and deployed 12+ real-world projects across Machine Learning, Full-Stack Development, and AI systems.
- Completed 20+ certifications spanning Machine Learning, Networking, Web Development, and System Design.
- Solved 150+ Data Structures and Algorithms problems across LeetCode and GeeksforGeeks, focusing on optimization and problem-solving patterns.
- Actively participated in technical projects, demonstrating rapid prototyping and problem-solving skills.

EDUCATION

Lovely Professional University — Phagwara, Punjab
Bachelor of Technology Computer Science and Engineering (ML) CGPA: 8.12 — Aug'23 – Present

Himgiri Public School — Panipat, Haryana
Intermediate (Class XII); Percentage: 74.83 — Apr' 21 – June'22

Himgiri Public School — Panipat, Haryana
Matriculation; Percentage: 78 — Apr'19 – Mar'20